

COLLINS IKIARA

Back-End Developer | Data Scientist

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PROFESSIONAL SUMMARY

Data Scientist and Back-End Developer with **5 years** of multidisciplinary experience in engineering, data analytics, and software development. Strong background in analytical problem-solving, machine learning, API development, and database architecture. Recently contributed to the development and launch of **Sowphie**, an Agri-Tech mobile application on Google Playstore, focusing on backend systems and database design.

CORE COMPETENCIES

1. **Languages & Frameworks:** Python, JavaScript, SQL, NoSQL, Version Control, REST APIs, FastAPI, Node.js, Express.js
2. **Tools:** Git, GitHub, Supabase, MySQL, MongoDB, Jupyter Notebooks, Power BI, Excel, VS Code
3. **Data Science:** Data Analysis, Predictive Analytics, Machine Learning, Data Modeling, Hypothesis Testing, Data Visualization, Descriptive Analytics
4. **Product Development:** Databases, API Development, API Testing, Database Administration, Product Research, Product Design, Prompt Engineering
5. **Development Operations:** DevOps, Backend Development, API Documentation, Database Architecture

PROFESSIONAL EXPERIENCE

Back-End Developer | SOWPHIE | April 2025 – January 2026

- Designed backend logic and REST APIs comprising approximately **140 endpoints** for the bookkeeping feature using **FastAPI** framework and prompt engineering
- Collaborated with **3-person development team** to build scalable backend infrastructure for agricultural management platform
- Co-designed database architecture deployed on **Supabase** as part of two-person team
- Led API testing efforts in collaboration with development team of 3 developers
- Refactored and organized backend codebase into modular structure including routers, models, service layers, and schema files
- Created and maintained comprehensive **API documentation** for all endpoints
- Contributed to successful launch of Sowphie app on **Google Playstore**

Data Scientist | OMDENA | August 2024 - October 2024

Project: The Impact of Climate Change on Maize Yield in Kenya

- Collaborated with cross-functional team of **10 data professionals** including recent graduates and PhD statisticians
- Conducted comprehensive climate trend analysis across **47 counties in Kenya**

- Performed yield analysis to assess climate impact on agricultural productivity
- Authored descriptive analytics documentation for research findings
- Applied statistical methods and data visualization techniques to communicate insights

Assistant Project Manager | ZED AXIS ARCHITECTS | February 2021 - April 2021

- Executed project directives to ensure daily deliverables were met on construction sites
- Managed technicians and subcontractors to maintain project timeline and quality standards
- Conducted daily and weekly material inventory management for optimal resource planning
- Prepared weekly progress reports and work schedules for client communication
- Enabled lead Project Manager to focus on strategic client engagement

Lead Graduate Engineer, Electrical Department | ZED AXIS ARCHITECTS | February 2018 - July 2019

- Designed electrical schematic drawings for diverse client projects and specifications
- Created technical plans and bill of quantities for contractor execution and cost estimation
- Collaborated with multidisciplinary consultants across Architecture, Landscape Architecture, Interior Design, Structural Engineering, and Mechanical Engineering
- Conducted electrical load analysis to advise clients on backup power solutions including **generators** and **solar energy systems**
- Delivered sound electrical designs aligned with client requirements across various project typologies

EDUCATION

6. [Data Science Certificate](#) | ALX (EXPLORE AI ACADEMY) | 2023 – 2024
7. [Foundations In Specialization \(FiS\)](#) | ALX ACADEMY | May 2023 – August 2023
8. [Bachelor of Science in Electrical and Electronic Engineering](#) | JOMO KENYATTA UNIVERSITY OF AGRICULTURE AND TECHNOLOGY | 2012 - 2016

CERTIFICATIONS

9. [Microsoft Azure AI Fundamentals \(Generative AI\)](#)
10. [Great Learning: Introduction to DevOps](#)
11. [freeCodeCamp: Python Certification](#)
12. [KodeKloud: KodeKloud Engineer – Git \(Level 1\)](#)
13. [Omdena: ML Engineer Certification](#)
14. [Strathmore University: CIPIT's Data Protection Course](#)
15. [Graduate Electrical Engineer](#)

PRODUCTS

Sowphie - Smart Farm Management Assistant

sowphie.io | [Google Playstore](#)

Agricultural mobile application serving farmers across Africa with **digital record-keeping** and **farm operations monitoring** for 100+ types of crops and livestock.

PROJECTS

Crop Recommender and Estimation System

- Developed machine learning-based crop recommendation system for smallholder farmers
- Integrated soil and environmental factors to provide crop suggestions
- Implemented harvest and revenue estimation features to support farmer decision-making
- Designed system to assist agricultural planners and policymakers

Crop Yield Predictor System

- Built machine learning system predicting crop yield for **22 different crop types**
- Utilized soil and environmental data as input features for prediction models
- Created tool to support farmers, agricultural planners, and policymakers in crop management decisions
- Enabled data-driven resource allocation and planning

World Layoffs Analysis (MySQL Project)

- Analyzed global company layoff data spanning **2020-2023**
- Applied SQL queries for data extraction, transformation, and analysis
- Generated insights on workforce reduction trends across industries and regions